In [1]:
```
pip install ipython-sql sqlalchemy
```

```
Requirement already satisfied: ipython-sql in c:\anaconda3\lib\site-packages (0.5.0)
Requirement already satisfied: sqlalchemy in c:\anaconda3\lib\site-packages (2.0.43)
Requirement already satisfied: prettytable in c:\anaconda3\lib\site-packages (from i
python-sql) (3.9.0)
Requirement already satisfied: ipython in c:\anaconda3\lib\site-packages (from ipyth
on-sql) (9.7.0)
Requirement already satisfied: sqlparse in c:\anaconda3\lib\site-packages (from ipyt
hon-sql) (0.5.5)
Requirement already satisfied: six in c:\anaconda3\lib\site-packages (from ipython-s
ql) (1.17.0)
Requirement already satisfied: ipython-genutils in c:\anaconda3\lib\site-packages (f
rom ipython-sql) (0.2.0)
Requirement already satisfied: greenlet>=1 in c:\anaconda3\lib\site-packages (from s
qlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\anaconda3\lib\site-pac
kages (from sqlalchemy) (4.15.0)
Requirement already satisfied: colorama>=0.4.4 in c:\anaconda3\lib\site-packages (fr
om ipython->ipython-sql) (0.4.6)
Requirement already satisfied: decorator>=4.3.2 in c:\anaconda3\lib\site-packages (f
rom ipython->ipython-sql) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in c:\anaconda3\lib\si
te-packages (from ipython->ipython-sql) (1.1.1)
Requirement already satisfied: jedi>=0.18.1 in c:\anaconda3\lib\site-packages (from
ipython->ipython-sql) (0.19.2)
Requirement already satisfied: matplotlib-inline>=0.1.5 in c:\anaconda3\lib\site-pac
kages (from ipython->ipython-sql) (0.2.1)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in c:\anaconda3\lib\sit
e-packages (from ipython->ipython-sql) (3.0.52)
Requirement already satisfied: pygments>=2.11.0 in c:\anaconda3\lib\site-packages (f
rom ipython->ipython-sql) (2.19.2)
Requirement already satisfied: stack_data>=0.6.0 in c:\anaconda3\lib\site-packages
(from ipython->ipython-sql) (0.6.3)
Requirement already satisfied: traitlets>=5.13.0 in c:\anaconda3\lib\site-packages
(from ipython->ipython-sql) (5.14.3)
Requirement already satisfied: wcwidth in c:\anaconda3\lib\site-packages (from promp
t_toolkit<3.1.0,>=3.0.41->ipython->ipython-sql) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in c:\anaconda3\lib\site-packages
(from jedi>=0.18.1->ipython->ipython-sql) (0.8.5)
Requirement already satisfied: executing>=1.2.0 in c:\anaconda3\lib\site-packages (f
rom stack_data>=0.6.0->ipython->ipython-sql) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in c:\anaconda3\lib\site-packages (f
rom stack_data>=0.6.0->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure_eval in c:\anaconda3\lib\site-packages (from sta
ck_data>=0.6.0->ipython->ipython-sql) (0.2.3)
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
%load_ext sql
```

In [3]:
```
%sql sqlite:///mydatabase.db
```

In [32]:
```
%sql mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&tru
```

In [5]:
```
%%sql
SELECT TOP 5 * FROM Superstore_Data;
```

```
 * mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
   sqlite:///mydatabase.db
Done.
```

Out[5]:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States

In [14]:
```
%%sql
SELECT top(10) City, round(SUM(Sales),0) as Total_Sales
FROM Superstore_Data
GROUP BY City
ORDER BY SUM(Sales) DESC;
```

```
 * mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
   sqlite:///mydatabase.db
Done.
```

Out[14]:

City	Total_Sales
New York City	256368.00
Los Angeles	175851.00
Seattle	119541.00
San Francisco	112669.00
Philadelphia	109077.00
Houston	64505.00
Chicago	48540.00
San Diego	47521.00
Jacksonville	44713.00
Springfield	43054.00

In [21]:

```
%%sql
SELECT TOP 10
    [Customer Name],
    COUNT([Order ID]) AS Tran_Num
FROM Superstore_Data
GROUP BY [Customer Name]
ORDER BY Tran_Num DESC;
```

```
 * mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
   sqlite:///mydatabase.db
Done.
```

Out[21]:

Customer Name	Tran_Num
William Brown	37
John Lee	34
Matt Abelman	34
Paul Prost	34
Chloris Kastensmidt	32
Edward Hooks	32
Jonathan Doherty	32
Seth Vernon	32
Arthur Prichep	31
Emily Phan	31

In [30]:

```
%%sql
create view Customers_Levels As
select [Customer Name],round(sum(Sales),0) as Total_Sales,
```

```
case
when sum(Sales)>10000 then 'VIP Customer'
when sum(Sales)>5000 then 'Medium Customer'
else 'Normal Customer'
end as Customer_Levels
from Superstore_Data
group by [Customer Name];
```

```
 * mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
   sqlite:///mydatabase.db
Done.
```

Out[30]: []

In [34]:

```
%%sql
select Customer_Levels,count(Customer_Levels) as Customer_Count
from Customers_Levels
group by Customer_Levels
order by count(Customer_Levels);
```

```
 * mssql+pyodbc://@localhost/Super_Store?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
   sqlite:///mydatabase.db
Done.
```

Out[34]:

Customer_Levels	Customer_Count
VIP Customer	20
Medium Customer	97
Normal Customer	677

In [ ]: